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Research Article

Online Intelligent Monitoring System and Key Technologies for Dam Operation Safety

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To realize the comprehensive intelligent upgrade of the Three Gorges Dam safety intelligent monitoring system (IMS), we focus on three core pillars real-time information processing, professional analytical evaluation, and digital management control systematically overcoming critical technical bottlenecks. By deeply integrating artificial intelligence (AI), Internet of Things (IOT), big data analysis, and geographic information system + building information modeling (GIS + BIM) ecosystems, we conducted a holistic diagnosis of existing monitoring systems to precisely identify operational pain points. Leveraging our proprietary innovations, including a GIS + BIM digital base, smart algorithm matrix, and BIM-based finite element computing system, we successfully developed the Three Gorges Dam intelligent monitoring platform, delivering five core value propositions: (1) Achieve real-time and historical aggregation of comprehensive data with dam safety management as the core, fully encompassing various types of environmental monitoring data. (2) Utilizing "GIS + BIM" as the technical foundation, construct a digital twin geometric model of the hub monitoring physical world, enabling intuitive and precise representation of engineering status. (3) Implement online rapid structural calculation, analysis, and early warning based on "BIM + Finite Element" technology, providing timely and reliable support for safety decision-making. (4) Establish a monitoring data analysis model through machine learning intelligent algorithms, deeply mining data value to enable intelligent prediction of potential safety hazards. (5) Promote digital transformation of manual inspection workflows using "IOT + Micro-INS" technology, enhancing inspection efficiency and accuracy. Additionally, our workflow engine ensures full-process digital collaboration across safety monitoring operations, guaranteeing seamless interdepartmental coordination. These innovations have not only enhanced safety management efficiency but also cemented the Three Gorges Dam's global leadership in hydraulic engineering. As a landmark achievement in national strategic infrastructure, it exemplifies the digital transformation of mega-scale engineering projects in the modern era.

Keywords: digital twin; intelligent monitoring system; machine learning intelligent algorithm; rapid structural calculation; safety monitoring

1. Introduction

Safety accidents are often the result of multiple factors working together. In the period of construction and operation, water conservancy and hydropower are easily influenced by long-term factors as environmental and load and sudden factors as floods and earthquakes disasters. In order to reduce the probability of hub accidents in water conservancy and

hydropower projects, practical and effective measures should be taken at all the stages of engineering design, construction, and operation. Safety monitoring as the "ears and eyes" of a hub, which can grasp the working and diagnose the health status of hub. Its significance lies not only in providing basic information for formulating early warning and emergency plans, providing engineering benefits and safety guarantees, but also in understanding the changing patterns of detection quantities to validate the rationality of design theories and methods. Thus, it can improve the scientific and technological level of hub engineering [1].

With the increasing scale of project, the operation and management of water conservancy and hydropower projects are becoming more and more complex, and all of these features bring more requirements and higher standards for safety monitoring. Traditional manual monitoring cannot make quick and effective analyses and decisions on numerous data, thus cannot meet the requirements of near real-time safety monitoring for water conservancy and hydropower projects. Therefore, keeping improving the safety management system, enhancing safety monitoring technology, and establishing accurate data analysis model, prediction models, and risk warning systems are inevitable. All of these works construct automation, informatization, and intelligence of safety monitoring that are necessary to ensure the efficient and reliable operation of the water conservancy and hydropower projects [2]. In recent years, significant achievements have been made in the automatic and informationalized construction, which lays a solid foundation for the construction of smart water conservancy. The rapidly developing digital twin technology integrates building information modeling (BIM), geographic information system (GIS), Internet of Things (IOT), artificial intelligence (AI), and other technologies and has a good application prospect in the field of water engineering safety monitoring, which can effectively improve the accuracy, timeliness, and integrity of data acquisition [3]. GIS technology has powerful functions of spatial data collection, storage, management, display, query, statistics, and description, which can accurately analyze data and provide strong support for auxiliary decision making [4]. BIM technology is an emerging design technology, with excellent 3D model rendering ability and data integration ability, with the advantages of visualization, quantification, real time, and sharing [5]. By using advanced technologies such as GIS, BIM, and the IOT, building a BIM + GIS engineering safety monitoring and early warning system can realize functions such as safety monitoring data management, two–three-dimensional integrated display of results, comprehensive monitoring analysis, and early warning management, which can greatly improve the level of engineering safety monitoring and early warning [6]. However, the increasing scale of the projects has led to various types of points for safety monitoring. Therefore, it is urgent to establish a scientific and unified online intelligent monitoring system (IMS), implement real-time collection and preprocessing of monitoring data, conduct in-depth online data analysis, and timely operation of Hub safety evaluation and early warning. This IMS can not only improve the informatization level of Hub safety monitoring and management and achieve accurate and rapid monitoring of the safety status of hydraulic structures but also provide timely information on the operation status of water conservancy and hydropower projects. Meanwhile, it is of great significance for enhancing the command and scheduling of flood control in the reservoir and even the basin, and thereby achieving the goal of reducing personnel and increasing efficiency, and avoiding the losses to the lives and property of millions and millions of people downstream caused by disasters such as Hub breaks [7, 8].

Currently, with the development of informatization, there still exists a certain gap between the informatization of safety monitoring in hydropower engineering and other industries. The gap is mainly manifested in obvious shortcomings in informatization and insufficient management capabilities, difficulty in sharing information, and low efficiency in development. The integration of management business and information technology is not close enough, making it difficult to fully leverage the overall advantages. Therefore, it is imperative to build an automated and information-based online IMS to comprehensively improve the safety management level of the dam operation. Therefore, there is an urgent need to establish an online IMS to comprehensively enhance the safety management level of the Three Gorges Hub. This will achieve an evolution from automation and informatization to intelligentization.

2. Automated Safety Monitoring System

The entire IMS for the operational safety of the Three Gorges project consists of three interconnected components: network architecture and system integration, data acquisition and transmission, and data processing and storage. These components collaborate synergistically to ensure fully automated operation of the system.

2.1. Network Architecture and System Integration

1. Network architecture: Based on the characteristics of the safety monitoring automation system (SMAS) for the dam operation, the SMAS is divided into three levels: monitoring center station, network management unit (NMU), and data acquisition unit (DAU). It is connected to the main fiber optic network of the dam operation, forming a hybrid network consisting of a fiber optic ethernet and multiple RS485 ring networks as in Figure 1.

The main function of the monitoring center station is to issue relevant instructions to the on-site NMU through the system network, automatically receive monitoring data at pre-set time intervals, and store it uniformly in the database according to the specified format. It can be viewed, analyzed, and reported through system software. The NMU and DAU are connected according twisted pair cables to form a circular field network with redundant communication bus lines. The RS485 communication protocol is used, which has strong anti-interference characteristics and robustness.

The NMU is an automation device for automated network management of security monitoring automation on-site networks and DAUs. The NMU has an automatic information

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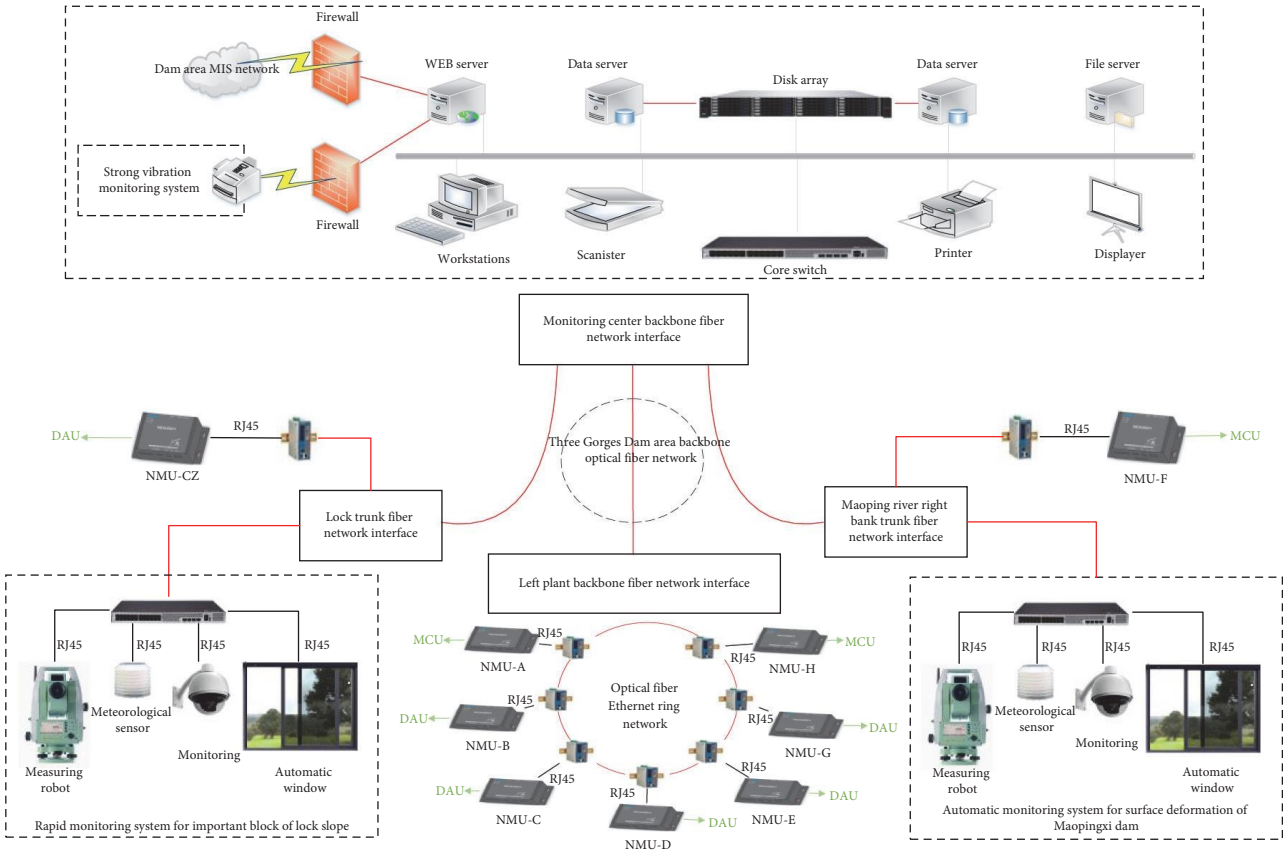


FIGURE 1: Network topology diagram of the security monitoring system.

distribution function, which can send instructions to the DAU under its jurisdiction for receiving, storing, and reporting data. The DAU can automatically collect and monitor sensor data. It can execute the collection instructions issued by NMU and can also automatically collect and store data according to its internal collection cycle.

2. System integration: In addition to connecting the internal monitoring sensors of various parts such as dams, ship locks, ship elevators, and Maopingxi, the system also has access to the surface deformation automation monitoring subsystem, strong vibration monitoring subsystem, and water and rain monitoring sub-system. Through switches and data integration interfaces, various subsystems are organically integrated together.

2.2. Data Collection and Transmission

1. Internal sensor data collection and transmission: DAU can perform periodic observations, while the system configures data collection cycles and times are set through security monitoring data collection and management software. After collecting data, it is first cached in the DAU and then uploaded to the higher-level NMU, and then, the NMU transforms it to the data server of the monitoring center station for storage.
2. Intelligent data collection and transmission of measurement robots: the self-developed observation software for measurement robots can intelligently observe based on the set cycles, meteorological conditions, or conditions of the deformation body. Observation data is automatically uploaded in real time, and the observation software can intelligently determine various tolerance indicators to perform re-measurement or observations storage operations.
3. Strong motion monitoring data collection and transmission: real-time seismic data is obtained through a strong motion instrument. When the intensity exceeds the threshold, the corresponding encrypted collection plan is activated, and the collected data is processed and analyzed before being stored in the database.

2.3. Data Processing and Storage

1. Data processing: by the data processing and calculation module for calculation and analysis, the measurement points that are out of limit or missing are recollected, then the observation and results that meet the requirements are stored in the compilation library and management by the information management system. The data processing flowchart is shown in Figure 2.

In the process of intelligent data collection and transmission by measuring robots, the observation quality of the

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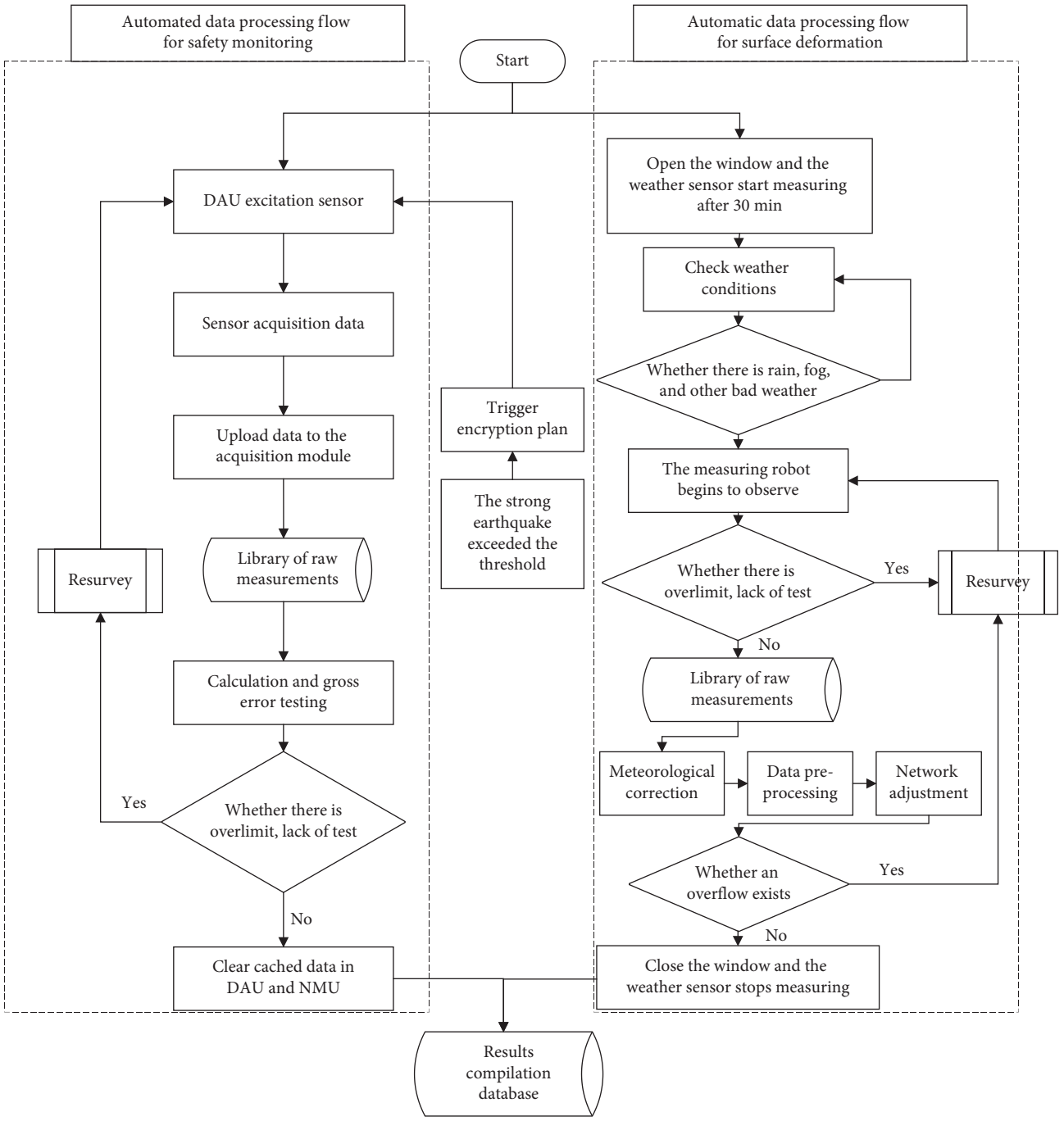


Figure 2: The data processing flowchart of security monitoring system.

measuring robot is easily affected by adverse weather conditions. Therefore, the opening and closing windows should be opened for more than 30 min before observation, and the observation can only be carried out after the meteorological conditions inside and outside are consistent. This strategy effectively ensures data quality and integrity, avoiding the occurrence of error warnings.

After obtaining qualified edge and angle observation data, the real-time network adjustment module begins to process the data, including meteorological correction, data process, network adjustment, accuracy evaluation, data verification, and other procedures. Then, the processed horizontal distance and horizontal angle are calculated using least squares adjustment to determine the coordinates and related parameters of each measurement point. Finally, the cumulative coordinates with gross error detection will be calculated and stored in the database [9].

2. Data storage: The monitoring center station has been deployed with two data servers by Raid10 which is an independent disk redundant array and shares a disk array to achieve dual machine hot backup to ensure

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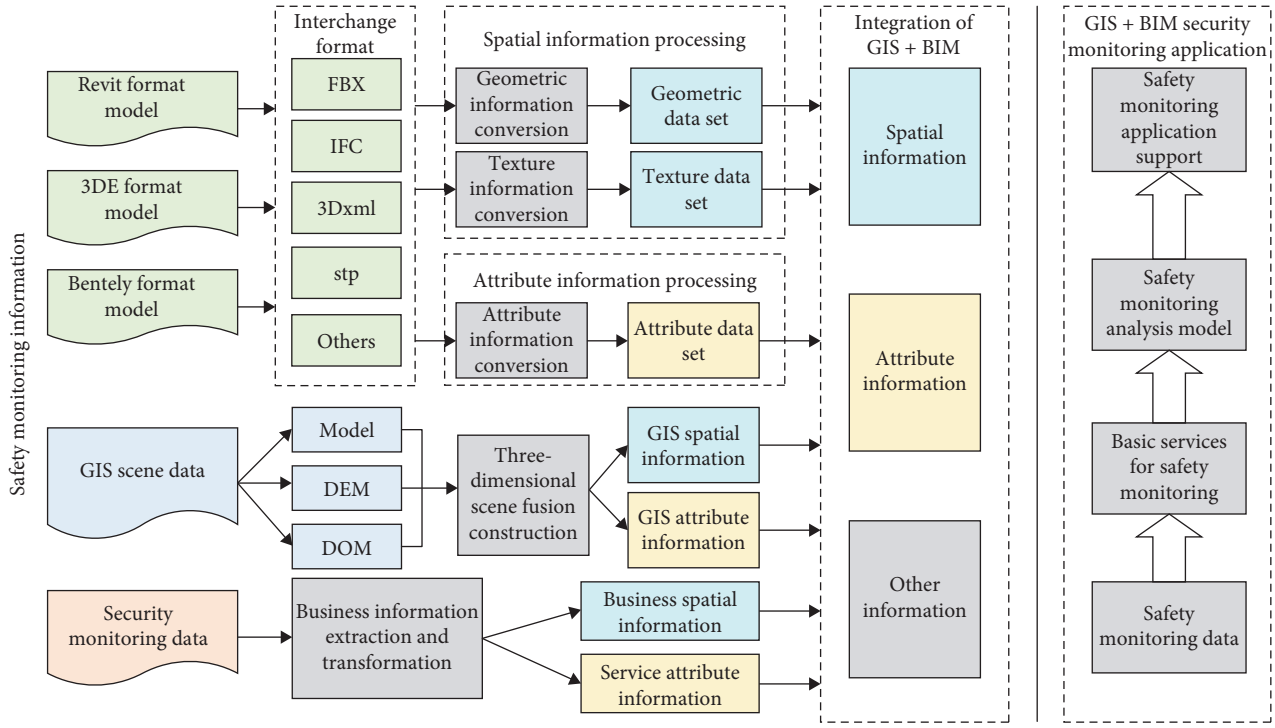


FIGURE 3: GIS + BIM data integration framework. BIM, building information modeling; GIS, geographic information system.

the safety of data upload and storage. Meanwhile, the dataset is used to achieve stripe set mirroring, providing 200% speed and data security against single disk damage.

3. Key Technology

The Three Gorges project operational safety online intelligent monitoring platform deeply integrates cutting-edge enabling technologies such as AI, IOT, big data, and GIS + BIM. It has formed five core technical systems and platform functional architectures, including GIS + BIM platform functional system, IOT + Micro-INS intelligent inspection technology, visualization technology, machine learning-based intelligent algorithm models, BIM, and finite element-based rapid structural calculation technology.

3.1. GIS + BIM Platform Construction. Aimed at the management needs of water conservancy and hydropower projects, a GIS + BIM platform has been constructed to solve problems such as massive data integration and scheduling and seamless integration of hydraulic buildings and 3D terrain. BIM model integration and fusion have achieved indoor and outdoor integration and display analysis of safety monitoring information, model-based and parameterized flood inundation analysis, real-time animation simulation, and other functions.

In terms of data integration, a BIM model transformation and integration framework for security monitoring GIS scenes is proposed, which completes the transformation and integration of commonly used BIM platform data models such as Revit, 3DE, and Bentley to security monitoring GIS scenes as shown in Figure 3.

In the aspect of model lightweight and shared scheduling, this work considers the characteristics of security monitoring scenarios and proposes a lightweight and shared scheduling framework for security monitoring scenarios using the level of detail (LOD) model and model sharing mechanism of GIS and BIM data. As shown in Figure 4, this framework can achieve efficient rendering and dynamic display of scenes at different scales and granularities. In addition, through data exchange, attribute integration, geometric lightweight, and other technical means, this article seamlessly integrates BIM data with 3D GIS, providing a scientific decision-making platform for visualization and analysis of safety monitoring.

3.2. Intelligent Inspection Technology. As shown in Figure 5, the intelligent inspection consists of a QR code, a Micro-INS module, an inspection terminal APP, and an online monitoring system WEB management module. A hybrid mode combining Client/Server (C/S) and Browser/Server (B/S) is adopted, which applies "inertial navigation of Micro+IOT" technology based on traditional manual inspection to ensure that inspection tasks are complete and abnormal parts are accurately located. Intelligent functions such as intelligent switching of positioning signals, inspection check-in, inspection records, inspection levels, inspection reports, and report generation are implemented.

1. For QR code labels: the positioning QR code is used in the corridor to provide positioning reference for Micro-INS when there is no GNSS or other information. The QR code of instruments and equipment mainly includes comprehensive information such as measurement point, instrument and equipment number, and burial and

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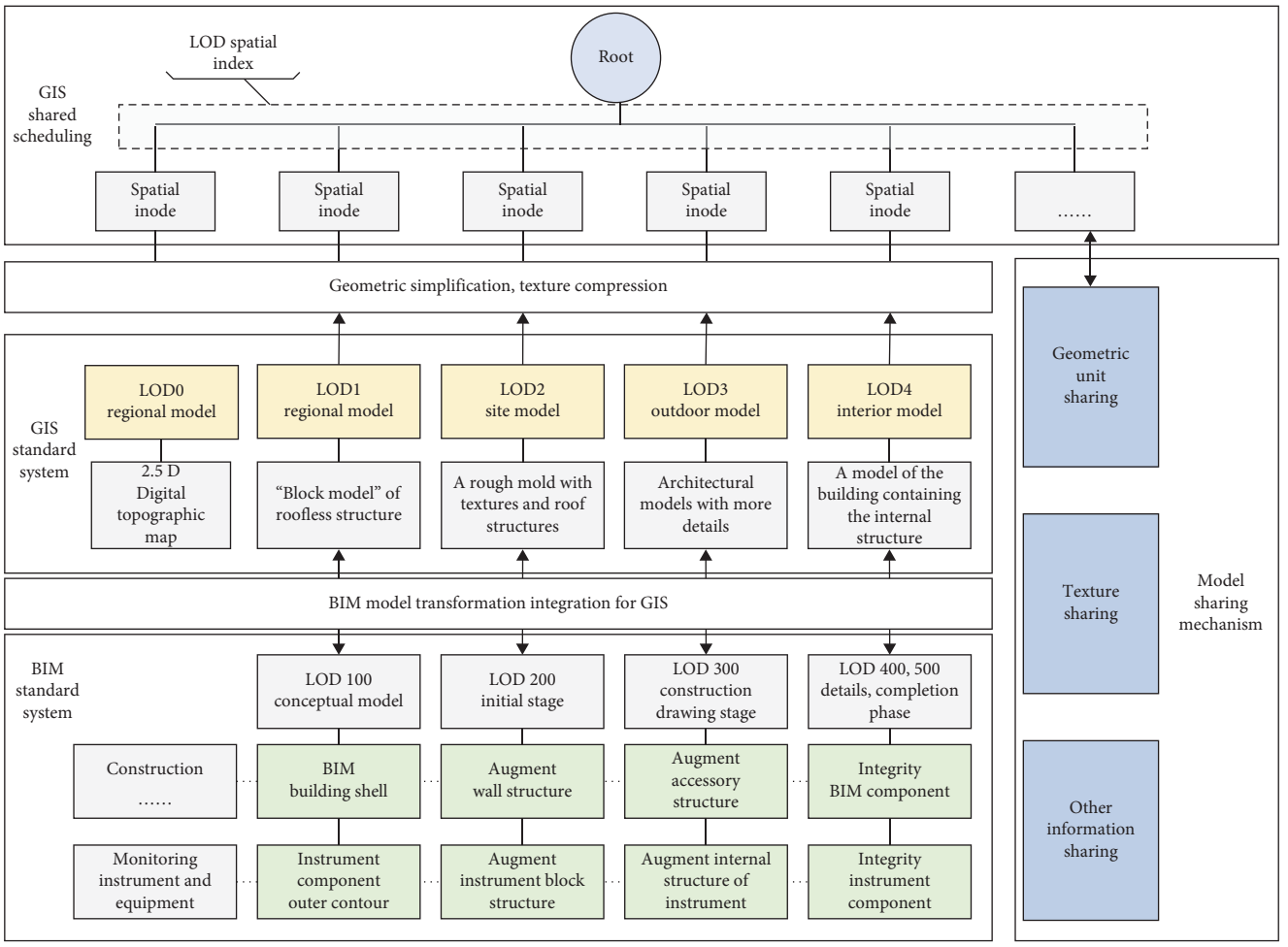


Figure 4: Model lightweight and shared scheduling framework.

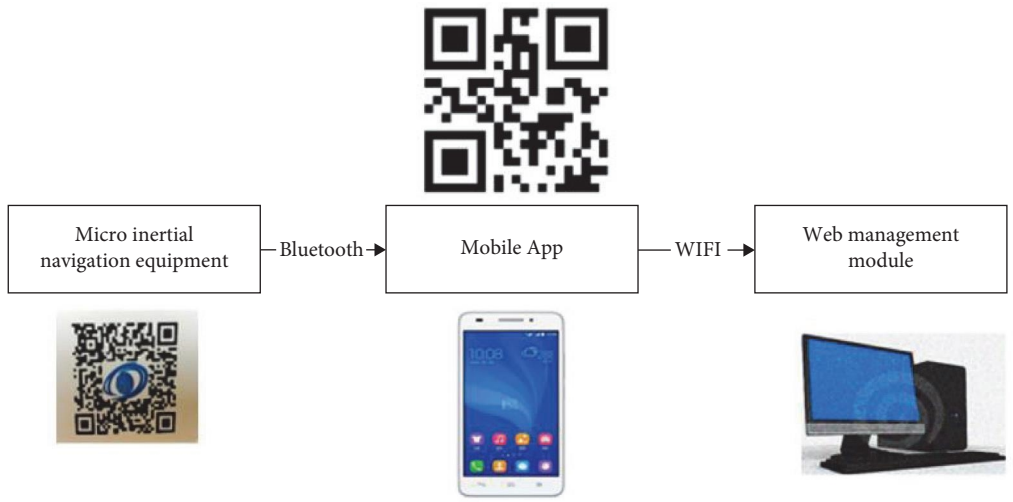


Figure 5: Intelligent inspection structure diagram.

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installation time, etc. The check-in QR code is mainly used for inspectors to clock in and record the on-site inspection trajectory. The abnormal point QR code mainly records information such as engineering diseases.

2. The main function of the Micro-INS module is to obtain the real-time coordinates of inspection personnel through the relative positioning algorithm of the INS module. The second is to transmit the coordinate and trajectory data from the inertial positioning module to the smartphone terminal through the bluetooth transmission function. Thirdly, the inspection trajectory is optimized through the optimization route algorithm to solve the problem of inspection trajectory deviation.
3. The main functions of the inspection terminal APP include permission management, task management, task execution, information management, data uploading, knowledge window, scanning, environmental quantity, equipment management, and auxiliary functions.
4. The main functions of the WEB management module include personnel management, department management, task management, check-in location management, comprehensive information management, and building hierarchy management.

3.3. Visualization Technology

1. Realistic 3D model: The system visualization is based on Cesium, on which various visualization technologies are organically integrated and applied. There is no browser dependency when using Cesium as the client, supporting changes to the JS source code and shader. In terms of GIS, it supports OGC standard services and various universal 3D data, and we proposes the only 3D model slicing format among current universal formats that supports streaming transmission and massive rendering of a large amount of geographic 3D data and 3D Files. By using this technology, it is possible to display and browse tilted real-life 3D models, BIM, and other 3D graphic elements in the system.
2. Multi–dimensional interaction of monitoring information: With 3D digitization and visualization technology, a virtual scene of hub visualization objects is constructed based on physical shooting, laser scanning, equipment drawings, videos, and other data. Multidimensional visualization of monitoring information enables the integration of monitoring data and 3D virtual reality, integrating the graphical calculation of 3D scenes and the numerical calculation of data analysis within the same architecture. It achieves the function such as scene visualization, status visualization, warning visualization, inspection visualization, and scheme visualization for safety monitoring as in Figure 6, providing collaborative data visualization analysis, and completing parallel processing of data mining, data mapping, 3D modeling, and visualization management.

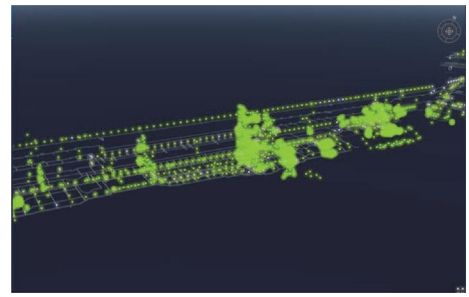


Figure 6: Visualization of monitoring status.

3.4. Intelligent Algorithm Model

1. A unified data preprocessing module suitable for all models has been developed based on the characteristics of dam safety monitoring data and the data requirements of analysis models. Its core technology is a factor processing program based on expression parsing, which defines the data processing methods of each factor through mathematical expressions. The program will parse the expressions and perform relevant preprocessing operations, which has strong universality and scalability.
2. Due to the complex working conditions and numerous influencing factors of actual dams, a single analysis model is insufficient in terms of stability. Thus, genetic algorithms were integrated into grey model (GM), back-propagation neural networks (BPNNs), and support vector machine (SVM) for parameter optimization. By employing an optimal weighted combination methodology, hybrid models such as GM–BP–SVM and statistical–BP–SVM were developed, resulting in substantially enhanced model performance [10–13].
3. A combined model of trend variation and periodic fluctuation was established to address the characteristics of trend variability and periodic fluctuation in dam monitoring time series. The model first uses trending decomposition algorithm to decompose complex binary time series into trend, periodic, and residual terms. Then, the characteristics of each sub item are analyzed, which can better conform to the characteristics of the double trending series, and is better than the single model [14, 15].
4. Due to the influence of reservoir capacity, water pressure, temperature, aging, and other uncertain factors on dams, resulting in highly nonlinear and nonstationary monitoring data of dams. Using traditional mathematical models to describe the relationship between these factors and dam monitoring data often has problems of insufficient accuracy and stability. In order to solve the above issues, an ensemble empirical mode decomposition (EEMD)–SVM–ARMA combination model has been has established. The combination model introduces modern mathematical methods such as set EEMD, SVM, and ARMA. This mode adopts EEMD which can help to overcome the high nonlinearity and nonstationarity of dam monitoring data, reduce data

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Table 1: Overall modeling accuracy of EEMD–SVM–ARMA combination model.

Process	Coefficient of association	Determination coefficient	Estimate standard error
Training	0.979205	0.958812	1.141762
Prediction	0.992126	0.981166	0.764442

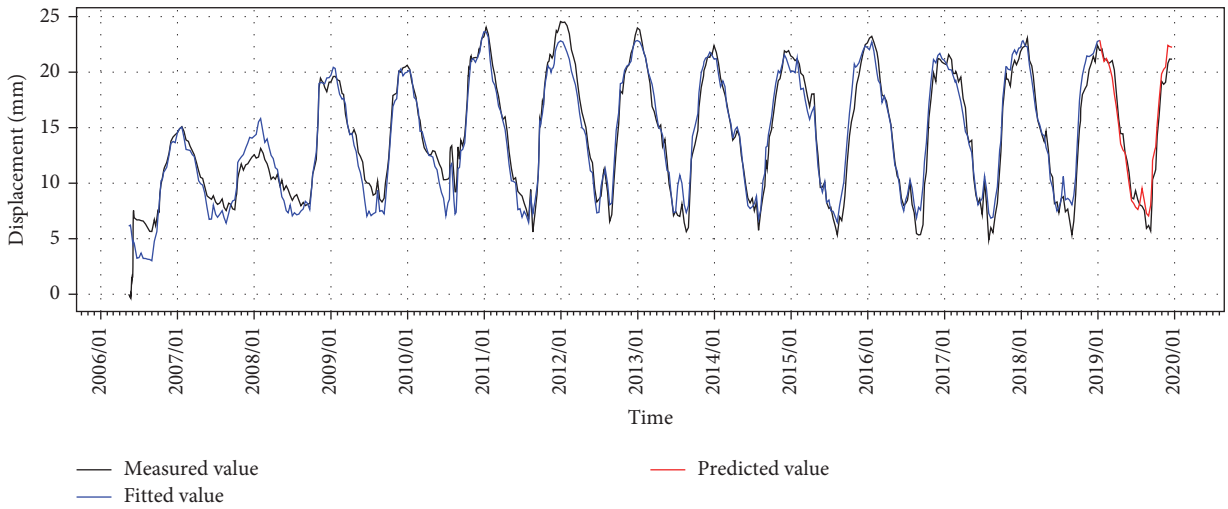


Figure 7: The calculated and measured values of EEMD–SVM–ARMA combination model.

complexity, and combine SVM and ARMA to further improve modeling and prediction performance. Meanwhile, the program adopts the fine-to-coarse automatically determine the high-frequency and low-frequency components, as well as the Hyndman Khandakar method to automatically determine the optimal ARMA model parameters, without manual parameter tuning work, and has high intelligence [16–20].

① First, we need to extract safety monitoring data: mainly including effect size data (such as displacement) and cause size data (such as water level, temperature, aging, etc.).

② We pre-process the causative data and calculated the water pressure, temperature and aging components: the water pressure component can be considered as a factor from the first to fifth order of water depth H, and the corresponding expression is as follows: $\sum_{i=1}^{3\sim5} a_{1i} H^i$

The corresponding expression for the temperature component: $\sum_{i=1}^{m_2} b_{2i} T_i$

The aging component usually has the following function types:

Polynomial: $\sum_{i=1}^{m} c_i t^i$;

Exponential function: $1 - e^{-c_1 t}$;

Logarithmic function: $\ln(1+t)$;

Hyperbolic function: $\frac{\xi_1 t}{\xi_2 + t}$;

Trigonometric function: $\sum_{i=1}^{2} (c_{1i} \sin \frac{2\pi i t}{365} + c_{2i} \cos \frac{2\pi i t}{365})$;

③ Then, we use a set of EEMD pairs of effect sizes to obtain a series of intrinsic mode functions (IMFs) and residuals, the decomposition formula is as follows:

$$x(t) = \sum_{i=1}^{n} \mathrm{IMF}_i(t) + r_n(t) \tag{1}$$

④ We use the fine-to-coarse algorithm to preprocess each IMF component and residual component to determine the frequency of IMF, which can be divided into high-frequency and low-frequency components.

⑤ For low-frequency components, we can use the Hyndman–Khandakar algorithm for automatic order determination, and establish the differential autoregressive moving average model (ARIMA) model. The calculation formula is as follows:

$$\begin{aligned} w_t &= \Delta^d x_t = (1-L)^d x_t \\ &= \phi_1 w_{t-1} + \phi_2 w_{t-2} + \mathrm{L} + \phi_p w_{t-p} + \delta + \mu_t + \theta_1 \mu_{t-1} \\ &\quad + \theta_2 \mu_{t-2} + \mathrm{L} + \theta_q \mu_{t-q} \end{aligned} \tag{2}$$

⑥ For high-frequency components, we use SVM for modeling and add the results of the high and low frequency calculations to get the final combined model calculation value.

The overall results of the training and prediction process of the EEMD-SVM-ARMA combination model are shown in Table 1 and Figure 7.

3.5. Fast Structural Calculation Technology Based on BIM and Finite Element Analysis. Based on the section of the Three Gorges Water Conservancy and Hydropower Project, a three-

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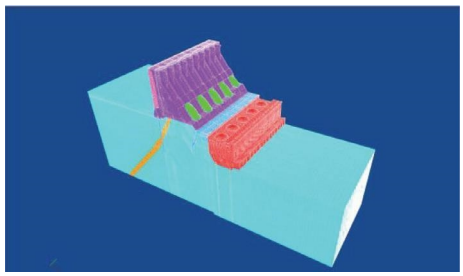


Figure 8: Web display effect of finite element mesh model.

dimensional and precise BIM model was established. Based on the Web-GL technology of the 5th generation HTML standard [21, 22], a complete set of functions such as interactive operation of structural finite element mesh model based on B/S architecture, static dynamic seepage/stress coupling and overall safety and stability analysis, 2D/3D all-round interactive display of calculation results, and feedback comparison of measurement points were achieved. It can provide the scientific basis for engineering and technical personnel to analyze and evaluate the safe operation of buildings. Taking into account factors such as the high maturity of large-scale commercial finite element software and the lack of open interfaces for preprocessing and post-processing, the rapid structural calculation analysis adopts a technical route of independent research and development of preprocessing and post-processing, as well as backend call for calculation and analysis. The development technologies for each functional module are as follows:

1. Finite element model interaction operation: the module is to call the finite element mesh model stored on the backend server to the web end for visual interactive display. The front-end requests the corresponding resource data from the backend service by calling the HTTP service interface published in the backend. The backend queries the database according to the received request parameters and returned result of JSON format. After receiving the interface data, the front-end uses Three.js for data visualization rendering and display. Figure 8 shows the web display effect of the 3D finite element mesh model for the dam section.
2. Static dynamic seepage, stress, stability analysis: the purpose of this module is to establish an HTTP communication protocol between the server and browser, transmit calculation instructions to the server, and automatically start relevant calculation work under the current login username. The implementation steps of finite element seepage field analysis are as follows:

According to Darcy's law of penetration, the penetration velocity in x, y, and z directions can be expressed as

$$\left.\begin{aligned} v_x &= -k_x \frac{\partial H}{\partial x} \\ v_y &= -k_y \frac{\partial H}{\partial y} \\ v_z &= -k_z \frac{\partial H}{\partial z} \end{aligned}\right\}. \tag{3}$$

Substitute the Formula (3) into the formula $\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$, we can get

$$\frac{\partial}{\partial x}\left(k_x \frac{\partial H}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial H}{\partial y}\right) + \frac{\partial}{\partial z}\left(k_z \frac{\partial H}{\partial z}\right) = 0. \tag{4}$$

For an isotropic seepage field, that is, when, Equation (4) becomes

$$\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} + \frac{\partial^2 H}{\partial z^2} = 0. \tag{5}$$

According to Equation (4), the finite element calculation formula for stable seepage flow can be written as

$$[K]\{h\} = \{F\}. \tag{6}$$

For steady seepage flow, the definite solution condition of the basic differential equation is only the boundary condition. The following categories are common:

- The first type of boundary condition or dirichlet condition:

$$H(x, y, z) = \phi(x, y, z)|_{(x,y,z)\in\Gamma_1} \tag{7}$$

- The second type of boundary condition or Neumann condition:

$$\vec{k} \cdot \left.\frac{\partial H}{\partial \vec{n}}\right|_{\Gamma_2} = q(x, y, z)|_{(x,y,z)\in\Gamma_2} \tag{8}$$

- Free surface and overflow surface boundary conditions:

$$\vec{k} \cdot \left.\frac{\partial H}{\partial \vec{n}}\right|_{\Gamma_3} = 0,\ H(x, y, z)|_{\Gamma_3} = z(x, y) \tag{9}$$

$$\vec{k} \cdot \left.\frac{\partial H}{\partial \vec{n}}\right|_{\Gamma_4} \neq 0,\ H(x, y, z)|_{\Gamma_4} = z(x, y). \tag{10}$$

This module implements multi-user online and multi-core shared memory parallel computing functions.

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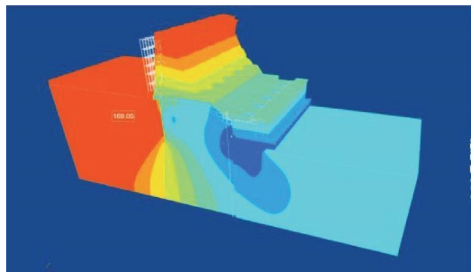


Figure 9: Interactive display effect of finite element postprocessing.

3. Result display and monitoring feedback comparison: the module is to load the finite element calculation results into the web for visual interactive display. The front-end page and functions of the module are written using Vue.js and Three.js, respectively. The interface style is determined through UI design. After reading the calculation results, the 2D/3D cloud map, vector, and measurement point monitoring feedback comparison are displayed in a comprehensive interaction manner, achieving automatic filling of tables and drawing of relevant curves. Figures 8 and 9 show the interactive display effect of 3D finite element postprocessing for the dam section.

4. Online Monitoring System Platform

4.1. System Architecture. The overall framework design of the system is layered according to the infrastructure layer, data layer, platform layer, and application layer. The overall design goal is to standardize development, modularize the system, containerized operations, containerized operation, and service-oriented applications. The overall architecture of the platform is shown in Figure 10.

1. Infrastructure layer: it is used to manage server computing resources, centralized and distributed storage resources, network resources, database resources, and security resources. Through virtualization, it realizes the management, expansion, and monitoring of software and hardware resources. With the use of horizontal expansion, it continuously supplements hardware resources to support the performance requirements of the upper platform layer and application layer.
2. Data layer: it is used for data/information collection as real-time collection, offline collection, internal/external network data collection, third-party data collection, and new data sources generated through data mining. The data storage it includes comprises structured data storage (SQL Server, Oracle, etc.), unstructured data storage (HBase, etc.), and spatial.
3. Platform layer: firstly, by abstracting the resources provided by the infrastructure layer, it allocates and shares resources in the form of resource pools, encapsulates applications using containers, and achieves rapid deployment and environment isolation. Then, by abstracting the distributed application model, it realizes fully automated application lifecycle management, covering functions such as software package management, multi-instance installation, network configuration, load balancing, fault recovery, monitoring, elastic scaling, and uninterrupted upgrades. Finally, through the micro-service framework, it abstracts the commonalities of business scenarios, centers around services, supports the requirements of multiple scenarios, and uses service governance to implement functions such as dependency management, security policies, operation and maintenance monitoring, service routing, traffic control, and efficient communication.
4. Application layer: realize multiterminal access to security monitoring information, as well as PC, web client, and mobile client, to achieve intuitive 2D/3D visualization. The users include four categories: safety monitoring implementation (monitoring implementation units, supervision units), safety monitoring business management, dam safety management, and system management and maintenance.

4.2. System Functions. Establishing an online intelligent monitoring platform to provide a collaborative work platform for dam safety monitoring units as implementation, supervision, dam operation safety management, and dam safety supervision and management, to improving the overall level and efficiency of dam safety management work. The main responsibility of the homepage display function of platform is to display monitoring statistical information and part key data. The homepage display is mainly divided into three categories: display version, management version, and business version. The functions of the IMS for the operation safety of the Three Gorges Dam are shown in the following Figure 11.

1. Measurement point information: includes three functional modules, data management, attribute management, and data query. The data management module realizes manual entry, offline batch import, temporary storage, reliability testing, and data entry process review. The attribute management module implements basic information management, grouping management, and layout management of measurement points. The data query module realizes the display of full measurement point data query, results statistics, process lines, distribution maps, correlation graphs, comparison of manual and automated data, and basic information statistics of measurement points.
2. Collection and control: automatically capture monitoring data through data interfaces. Support the release of single-point and batch collection commands to achieve online data collection, transmission, and information processing. It mainly includes functions such as point selection, group selection, plan selection measurement, and plan configuration.
3. Inspection management: using indoor inertial and outdoor GNSS positioning, a mobile intelligent inspection system is created by integrating smartphone terminals and web platforms via network communication. It has

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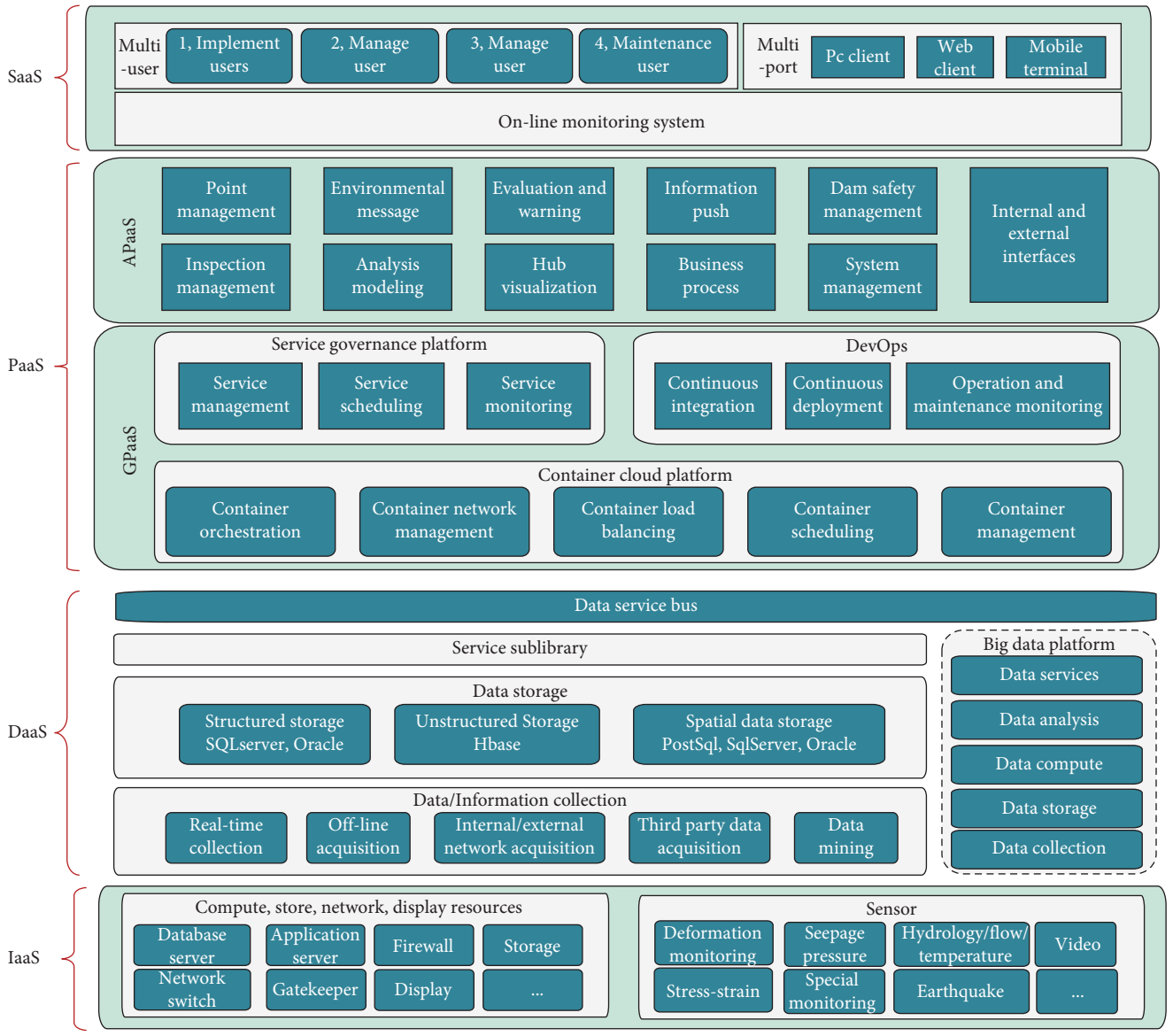


Figure 10: Overall architecture of online monitoring system.

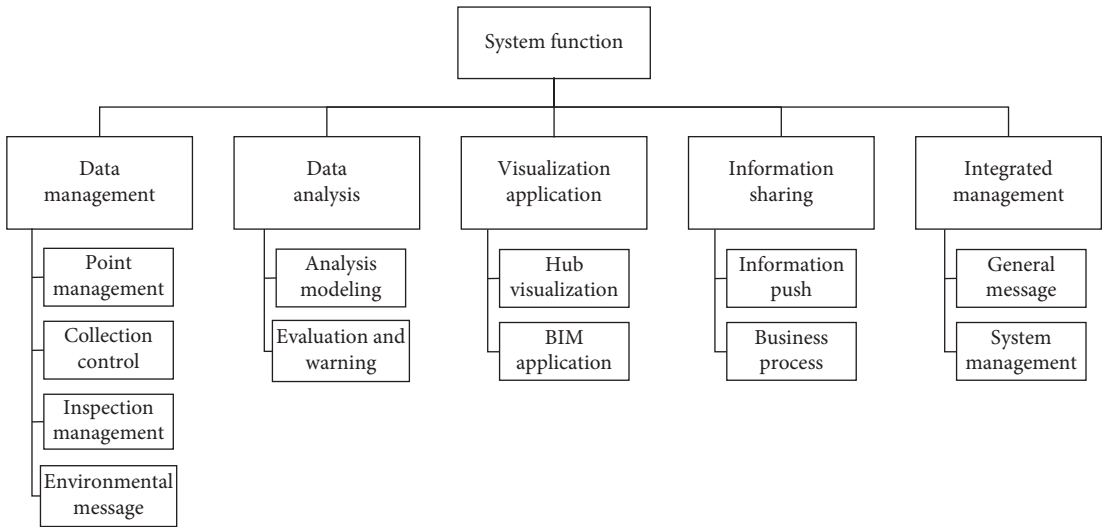


Figure 11: Functions of online monitoring system.

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functions like custom task setting, info push—reception, result analysis—report generation, and info storage—query, making inspection tasks more intelligent.

4. Environmental information: the environmental and operational characteristic information management module is responsible for managing the environmental information, earthquake monitoring information, and engineering characteristic information collected by the project. Environmental information mainly refers to hydrological and meteorological data (upstream and downstream water levels, inflow and outflow, temperature, precipitation, etc.), and engineering characteristic information includes floodgates, unit operation, ship lock operation, and water filling and discharging.
5. Analysis and modeling: this module mainly includes specialized calculations such as stress freed calculation analysis, strain gauge group calculation analysis, correlation calculation analysis, and infiltration line calculation analysis. Data organization includes data verification, data interpolation, and data preprocessing functions. Analysis and modeling include conventional mathematical models, intelligent algorithm models, model management, and update maintenance functions.
6. Evaluation and early warning: it includes achievement management, monitoring system evaluation, monitoring data evaluation, fault alarm response, data anomaly warning, and other contents.
7. Hub visualization: it integrates three core modules. the 3D Visualization Platform of the hub integrates three core modules. 3D display of basic geographic information which are high-definition images, oblique photography models, and real-scene views of the dam; 3D analysis of architectural structures which is spatial configurations and associated relationships of buildings; and dynamic presentation of monitoring data which are instrument layout, real-time status, data collection, and simulation analysis. This enables the digital mapping, monitoring, and management of all elements of the Three Gorges hub.
8. BIM application: Web-GL technology based on the 5th generation HTML standard has achieved online calculation of structural finite elements and antislip stability based on the B/S architecture, providing the scientific basis for engineering and technical personnel to analyze and evaluate the safety status of building operation.
9. Information push: including safety information push for hydropower dam operation—National Energy Administration Dam Safety Center; data push on the characteristics and operation status of watershed cascade engineering—national energy administration information center; comprehensive monitoring information push for operational safety—Yangtze River Water Resources Commission.
10. Business process: classify and manage various processes in the security monitoring business, including three submodules: initiating process, process pending, and completed process. The initiating process module categorizes all the processes that the current user can initiate, the pending process module categorizes all the processes that the current user needs to handle, and the completed processes categorize the processes that the current user has already processed. Users can check the flow progress of the processes in the completed processes.
11. Integrated management: it is mainly based on electronic archive documents, utilizing a distributed file system, combined with structured information management of relevant data, to achieve comprehensive management of dam operation safety. Adopting a document resource directory management method to achieve classification management and maintenance of various documents. Set various key attributes for filtering and querying, achieving full-text document retrieval.
12. System management: It is an auxiliary part of the online security monitoring system, which requires user and permission management, data backup and recovery, system operation status monitoring, and log management functions. Currently, management of user role permissions and data permissions for distributed submeasurement points have been provided including the display field attribute settings for viewing admission forms, data forms, process lines, etc. in the system, including the setting of default parameter symbols and formulas for measuring points, including functions such as querying system operation logs by users.

4.3. System Characteristics

1. The industry's first security monitoring information management platform is based on GIS + BIM. By leveraging the visualization carriers of GIS + BIM, it realizes the multidimensional and multielement integrated dynamic display of the dam area of the Three Gorges Project, as well as services such as structural calculation, analysis and modeling, and evaluation and early warning.
2. Online fast structural calculation based on BIM and finite element, directly switching from the BIM model browsing module to the finite element calculation module. When the external load conditions change, online analysis and evaluation of the operation status of the dam foundation system can be achieved.
3. Implementing security monitoring business process informatization based on workflow engines, utilizing industry-leading workflow engines and seamlessly integrating with online monitoring systems, supporting cross-platform single sign-on. Meet the customization and reengineering of security monitoring business processes and achieve paperless and information-based office work.

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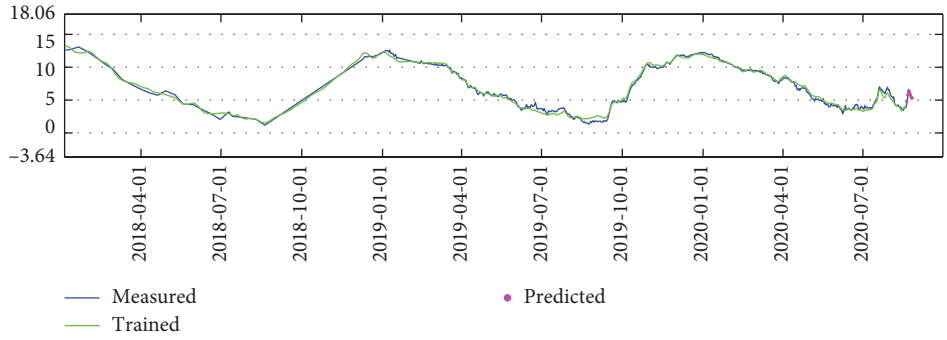


Figure 12: PL04ZC052 predicted displacement process line for Dam Section 5 of Left Factory.

Table 2: Statistical table of predicted deformation and displacement of left factory dam Section 5#.

Dam deformation displacement (mm)	Measured (2020.8.10)	Predicted (2020.8.20)	Predicted (2020.8.22)	Variation (8.20–8.10)	Variation (8.22–8.10)
Left factory 5# dam foundation	1.73	1.74	1.91	0.01	0.18
Left factory 5# dam top	4.03	4.84	6.36	0.81	2.33

4. To realize the integration of historical data, automatic observation of real-time online monitoring data, access to environmental and engineering operation characteristic information and the aggregation of comprehensive monitoring data. It offers conventional mathematical models and intelligent algorithm models, incorporating big data correlation analysis processing.
5. The system automatically completes security monitoring (automated and manual observation) operation and monitoring data analysis and evaluation. In case of abnormal operating environment information, the monitoring system can be triggered to automatical encrypt observations. When monitoring data anomalies in real time, it will achieve data anomaly warning and prediction.
6. Based on the intelligent inspection technology of "IOT+Micro-INS" to achieve the informatization of manual inspections. By using Micro-INS modules and optimal INS algorithms to realize intelligent functions such as no omission of inspection tasks, accurate positioning of abnormal parts and self-generation of inspection reports can be achieved.

5. Application of Dam Safety Performance Prediction Assisted in Flood Peak Discharge Flow Control Decision

The online monitoring system and key technologies studied in this article have been successfully applied in the stability and safety prediction analysis of dams under the condition of exceeding the standard flood in 2020, providing detailed, accurate, and intuitive support data for the control of discharge flow during the maximum flood peak period of the dam and generating huge benefits.

5.1. Prediction of Dam Behavior. On August 17, 2020, the flood control dispatch forecast showed that the peak flow of the Three Gorges reservoir reached 75,000 m^3/s on August 20. To successfully respond to this flood, a pre-exercise was conducted on the stability and safety of the dam under flood conditions, and a dam monitoring statistical analysis model and rapid structural calculation of the dam were initiated.

1. Prediction of statistical analysis model for dam monitoring: using a monitoring statistical analysis model to predict the deformation of the dam during this flood process, the flood monitoring statistical analysis model predicts the deformation displacement process of the 5th dam section of the left factory, as shown in Figure 12, and the predicted deformation displacement is shown in Table 2.

When the flood occurs on August 20, 2020, the forecast shows that the downstream displacement of the dam foundation of the 5th dam section of the left factory is 1.74 mm, and the downstream displacement of the dam crest is 4.84 mm. Compared with before the flood peak (August 10, 2020), the displacement change of the dam foundation is 0.01 mm, and the displacement change of the dam crest is 0.81 mm.

Forecast results after the flood peak on August 22: The downstream displacement of the dam foundation of the No. 5 dam section of the left factory is 1.91 mm, and the downstream displacement of the dam crest is 6.36 mm. Compared with before the flood peak (August 10, 2020), the displacement change of the dam foundation is 0.18 mm, and the displacement change of the dam crest is 2.33 mm.

The forecast results indicate that the displacement change has not exceeded the historical extreme and warning values, and there is no need to activate the dam safety emergency plan.

2. Online calculation and analysis preview of dams: by using the online calculation and analysis method of

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Table 3: Online calculation results before and after flood peak at the year of 2020.

Project	Dam block	Position	Before the flood (8.10)	At the peak of forecast flood (8.20)	After the forecast flood peak (8.22)	Variation (8.20–8.10)	Variation (8.22–8.10)
Displacement (mm)	Left factory 5#	Dam foundation	1.40	1.77	2.33	0.37	0.93
		Dam crest	4.62	6.12	8.56	1.50	3.94
Osmotic Pressure (m)	Left factory 3#	behind the white curtain	51.16	51.35	51.51	0.19	0.35
Stress (MPa)	Left factory 5#	Dam heel	−0.74	−0.69	−0.61	0.05	0.13

Note: The online calculation and prediction of the dam indicate that the Yangtze River 5# flood has a relatively small impact on the deformation, seepage, and stress of the dam.

Table 4: Dam deformation situation at the year of 2020 (unit: mm).

Dam block	Position	Pre-flood (08.10)	Pattern for wood over dam (8.20)	Upper level 167.65 (8.22)	Variation (8.20–8.10)	Variation (8.22–8.10)
Left factory 1#	Foundation ▼ 95.0	1.00	0.85	1.18	−0.15	0.18
	Dam crest	3.93	4.32	6.40	−0.19	0.28
Left factory 5#	Foundation ▼ 94.2	2.15	2.35	2.56	0.2	0.41
	Dam crest	3.68	4.40	6.35	0.72	2.67
Flood discharge 1#	Dam Crest	3.94	5.56	8.28	1.62	4.34
Flood discharge 2#	Foundation ▼15.1	1.91	1.57	1.45	−0.34	−0.46
	Dam crest	5.94	7.18	10.21	1.24	4.27

dams studied in this article, the dam behavior before the flood peak is calculated, the dam behavior when the flood peak passes is predicted, the dam behavior during the flood peak is evaluated, and the dam behavior after the flood peak is predicted. After calculation, the measured results are close to the forecast results. The online calculation results before and after the flood peak are shown in Table 3.

5.2. Actual Measurement Results. After the flood, the automation measurement results of the dam were compiled as follows.

The deformation of the dam before and after the flood peak is shown in Table 4. It can be seen from Table 4 as follows:

1. On August 10, 2020, the cumulative horizontal displacement of the dam foundation water flow was 1.00 mm (left factory 1) to 2.15 mm (left factory 5). Compared with the peak flood passing through the dam on August 20, the displacement change was −0.34 mm (flood discharge 2) to 0.2 mm (left factory 5); Compared to August 22nd, the displacement has changed from −0.46to 0.41 mm.
2. On August 10, 2020, the cumulative horizontal displacement of the water flow at the top of the dam was between 3.68 mm (left factory 5) and 5.94 mm (flood discharge 2). Compared with the peak flood passing through the dam on August 20, the displacement change was between −0.19 mm (left factory 1) and 1.62 mm (flood discharge 1); Compared to August 22, the displacement change is between 0.28 mm (left factory 2) and 4.34 mm (flood discharge 1).

5.3. Comparison and Analysis of Forecast and Actual Measurement Results. After the flood, the monitoring results of the dam were compiled, and a comparative analysis was conducted between the measured and predicted results. The comparative analysis was conducted between the forecast and actual measured results, as shown in Table 5. It can be seen that the measured results of the left factory dam section are consistent with the forecast and rehearsal results of this project. The maximum difference between the rehearsal and the measured value is within 1.3 mm, and the flood has little impact on the dam.

6. Conclusion

The operational management of water conservancy and hydropower projects in the modern era has grown increasingly complex alongside the expansion of project scales. Enhancing safety management and risk warning systems, improving technologies, and establishing precise analysis and prediction models are crucial for advancing automation, informatization, and intelligence of safety monitoring. These efforts are essential to ensure the efficient and reliable operation of water conservancy and hydropower projects. Thus, we have developed an online IMS platform for ensuring the operational safety of the Three Gorges Dam. This platform innovatively integrates

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Table 5: Comparison between forecast and actual results of water head variation (left factory 5# dam section displacement) unit: mm.

Position	Dam monitoring statistical analysis model (forecast)		Online computational analysis (forecast)		Measured results (measured)	
	162.0 m (8.20–8.10)	167.8 m (8.22–8.10)	162.0 m (8.20–8.10)	167.8 m (8.22–8.10)	161.81 m (8.20–8.10)	167.65 m (8.22–8.10)
Foundation	0.01	0.18	0.37	0.93	0.2	0.41
Dam Crest	0.81	2.33	1.50	3.94	0.72	2.67

technologies and innovation as follows: (1) Using GIS + BIM technology, the system seamlessly integrates GIS, BIM, monitoring data, and other resources into a unified spatial reference framework. This achieves comprehensive three-dimensional representations of the central project area, facilitating data collection and visualization across various scales from macro to micro, ground level to aerial views, indoors to outdoors, and span both 2D and 3D. (2) This intelligent system utilizing "INS + IOT" digital mapping achieves customized task management, field information collection, inspection track recording and replay, automatic report generation, and other functionalities, significantly enhancing the informatization level of safety inspection and monitoring at the Three Gorges project. (3) Through an optimal weighted combination algorithm, a composite model integrating multiple intelligent algorithms including BP model and the SVM model has been developed. It has enhanced both the stability and prediction accuracy of dam deformation model. (4) We integrate BIM, WebGL, finite element analysis, and online monitoring technologies. The system realizes full-process visualization of online finite element analysis and evaluation, providing a scientific basis for assessing the safety characteristics of buildings. (5) Drawing upon key physical parameters such as deformation and seepage in dam operation safety, a method for predicting the safety state of dam operations has been proposed. This effectively supports the functions of forecasting, early warning, simulation, and preplanning for ensuring dam operation safety. This online intelligent platform of the Three Gorges Dam significantly enhances the standardization, institutionalization, and regularization of operational safety information management and reporting for the watershed hub, thereby laying a robust foundation for the operational security monitoring of a smart watershed hub.

Data Availability Statement

The data used to support the findings of this study are included within the article.

Conflicts of Interest

The authors declare no conflicts of interest.

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